

Fusion draft: SST-23/56/63 Vacuum / dual medium cluster

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April 19, 2026

About this fusion draft

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Chapter 1

SST-23 — Dual vacuum unification

Swirl Superradiance, Photon Torsion, and the Unruh Echo Mechanism

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Abstract

We develop a falsifiable hydrodynamic extension of field theory in which the vacuum consists of two interacting, incompressible continua: a fast electromagnetic sector with characteristic propagation speed c , and a slower rotational Swirl sector with speed $|\mathbf{v}_\zeta| \sim 10^6$ m/s. Using Helmholtz–Kelvin vortex dynamics, we show that acceleration excites vorticity stretching rather than compression, producing a primary Swirl burst with characteristic delay $\tau_S \sim 0.1$ ns. Because electromagnetic cavities suppress shear-wave impedance by $\kappa_{se} \sim 10^{-7}$, this burst dissipates non-radiatively, leaving a secondary photon “Unruh Echo” at ~ 30 ns—the signal observed in recent cavity superradiance experiments. We formulate a coupled-rate theory of Swirl–EM energy flow, introduce a torsion-modified Maxwell equation, derive a parity-odd linear-in- B vacuum birefringence, and propose a BEC vortex-lattice experiment capable of distinguishing this dual-vacuum picture from standard quantum field theory.

Contents

1.1 Introduction

Quantum Field Theory (QFT) and General Relativity (GR) model the vacuum as a single Lorentz-invariant medium defined by a fixed causal speed c . In contrast, the Swirl String Theory (SST) framework proposes that the vacuum comprises two interacting, incompressible substrates:

1. A fast electromagnetic (EM) sector with characteristic wave speed c .
2. A slower hydrodynamic Swirl sector with rotational wave speed $|\mathbf{v}_\Omega| \sim 10^6$ m/s.

This “Dual-Vacuum Hypothesis” naturally resolves an outstanding puzzle: recent cavity experiments have observed superradiant delay times $\tau_d \sim 30\text{--}50$ ns consistent with standard QFT predictions for the Unruh effect [1, 2], while providing no evidence of non-standard vacuum excitations.

We argue that these experiments probe only the photon echo of a deeper, faster, hydrodynamic event. Because the Swirl sector supports shear/twist modes rather than compressional modes,

$$\nabla \cdot \mathbf{v} = 0, \quad \rho_f = \text{const.}, \quad (1.1)$$

acceleration excites *vorticity stretching* rather than pressure waves. The characteristic Unruh-like temperature of this excitation is amplified by the velocity ratio

$$\eta = \frac{c}{|\mathbf{v}_\Omega|} \approx 274. \quad (1.2)$$

Thus the Swirl sector responds ~ 274 times faster than the EM sector, producing a sub-nanosecond torsional “Unruh precursor.” Because the cavity impedance is mismatched for shear waves by $\kappa_{se} \sim 10^{-7}$, this precursor is unobservable, leaving only a secondary photon outburst—the “Unruh Echo.”

In this article we (i) formalize this hydrodynamic interpretation, (ii) derive the coupled Swirl–EM rate equations, (iii) introduce a torsion-modified Faraday law, and (iv) propose a vortex-lattice BEC experiment capable of falsifying the model.

1.2 Historical Context: Helmholtz, Kelvin, and Unruh

Helmholtz [4] showed that vorticity is materially conserved in an inviscid, incompressible fluid. Kelvin [5] extended this to vortex filaments, demonstrating that their topology and circulation Γ remain invariant. These foundations underpin modern superfluid dynamics and motivate SST’s incompressible-vacuum framework.

Unruh’s 1976 prediction [3] that acceleration induces thermal vacuum noise is typically derived assuming a single, massless field propagating at c . Recent theoretical proposals [1, 2] have shown that cavity-enhanced superradiance can amplify this effect into an observable photon burst.

Our work unifies these ideas: the Unruh effect is reinterpreted as a vorticity-stretching instability in an incompressible rotational substrate, and photon emission is a secondary transduction event.

1.3 Vorticity Intensification Under Acceleration

In incompressible Euler flow, vorticity evolves by the stretching equation:

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{v}. \quad (1.3)$$

Acceleration of an embedded vortex filament produces background shear, which stretches the filament, reducing its core area A_c while preserving circulation,

$$\Gamma = \int \boldsymbol{\omega} \cdot d\mathbf{S} = \omega A_c = \text{const}. \quad (1.4)$$

Thus $|\boldsymbol{\omega}|$ increases as A_c decreases. The local swirl energy density rises,

$$\rho_E = \frac{1}{2} \rho_f |\mathbf{v}|^2, \quad (1.5)$$

without violating incompressibility.

This intensification is the hydrodynamic analogue of “Unruh excitation” in standard QFT. Because the Swirl sector’s characteristic wave speed is $|\mathbf{v}_\odot| \ll c$, the effective Unruh temperature becomes

$$T_{\text{SST}} = \frac{\hbar a}{2\pi k_B |\mathbf{v}_\odot|} = \eta T_U. \quad (1.6)$$

Vorticity amplification therefore occurs on timescales

$$\tau_S \sim \frac{|\mathbf{v}_\odot|}{c} \tau_\gamma \approx 0.1 \text{ ns} \quad (1.7)$$

for photon-burst delays $\tau_\gamma \sim 30 \text{ ns}$ observed in cavity experiments.

1.4 Dual-Vacuum Architecture: Swirl and EM Sectors

The Swirl sector supports torsional (Kelvin-like) waves, while the EM sector supports transverse electromagnetic waves. Their impedances are

$$Z_S = \rho_f |\mathbf{v}_\odot|, \quad Z_\gamma = Z_{\text{bound}} \sim 10^7 \text{ Rayl}. \quad (1.8)$$

The Swirl–EM transduction coefficient is

$$\kappa_{se} = \frac{4Z_S}{Z_{\text{bound}}} \sim 10^{-7}, \quad (1.9)$$

so only one part in 10^7 of Swirl energy couples to photons.

This explains why cavity experiments detect only the late electromagnetic burst and not the fast Swirl precursor.

1.5 Torsion-Modified Electrodynamics

The electromagnetic field responds to time-varying vorticity via an additional torsional source term. The modified Faraday law is

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} - \mathcal{G} \partial_t (\nabla \times \mathbf{v}) \hat{n}, \quad (1.10)$$

where $\mathcal{G}$ is a coupling constant and $\hat{n}$ is the swirl-string director. This produces:

- (i) A linear-in- B vacuum birefringence, distinct from QED’s quadratic response.
- (ii) Parity-odd optical activity due to the chiral swirl-clock.

1.6 Coupled Swirl–Photon Rate Dynamics

Let n_S and n_{EM} be Swirl and EM excitation populations. Their evolution is

$$\dot{n}_S = -(\Gamma_S + \gamma_{\text{diss}} + \kappa_{se}) n_S, \quad (1.11)$$

$$\dot{n}_{EM} = \kappa_{se} n_S - \Gamma_{EM} n_{EM}. \quad (1.12)$$

Here

- $\Gamma_S = \eta \Gamma_{GR}$ is the Swirl superradiant rate.
- Γ_{EM} is the cavity-enhanced photon decay rate.
- γ_{diss} encodes non-radiative Swirl damping.

Solving yields

$$n_{EM}(t) = \frac{\kappa_{se}}{\Gamma_{EM} - \Gamma'_S} \left(e^{-\Gamma'_S t} - e^{-\Gamma_{EM} t} \right), \quad (1.13)$$

where $\Gamma'_S = \Gamma_S + \kappa_{se} + \gamma_{\text{diss}}$.

The rising edge forms a “prethermalization plateau,” while the EM peak at $t \sim 1/\Gamma_{EM} \approx 30$ ns matches experimental data [1].

1.7 Experimental Predictions

1. **Dual-burst emission.** A Swirl precursor at ~ 0.1 ns followed by a photon echo at ~ 30 ns.
2. **Impedance sensitivity.** Echo amplitude scales as $\kappa_{se} \propto Z_S/Z_{\text{bound}}$.
3. **Vacuum birefringence.** Linear-in- B optical rotation due to torsion term (1.10).
4. **Medium dependence.** Unruh temperature scales as $T_{\text{SST}} \propto 1/|\mathbf{v}_\mathcal{O}|$.
5. **Absence of density waves.** No acoustic emission whatsoever; signal is purely vorticity-based.

1.8 Falsification Experiment: Accelerated Vortex-Lattice BEC

A vortex-lattice Bose–Einstein condensate responds to torsional disturbances through Kelvin modes [7]. Accelerating such a lattice should excite a fast torsional precursor. SST predicts a sharp Kelvin-mode pulse at ~ 0.1 ns; QFT predicts no such pulse.

A null detection rules out hydrodynamic dual-vacuum theories. A positive detection falsifies the standard view of vacuum structure.

1.9 The SST Rosetta Stone Summary Table

1.10 Conclusion

We have presented a hydrodynamic dual-vacuum theory unifying Swirl and EM excitations through vorticity stretching, cavity impedance, and torsion-modified Maxwell dynamics. This framework explains the timing and shape of Unruh-superradiant signals observed in accelerated cavities and predicts a fast, currently undetected Swirl precursor. The proposed BEC vortex-lattice experiment provides a clear, falsifiable test.

Table 1.1: The SST Rosetta Stone: Mapping Fluid Variables to Standard Physics

Standard Constant	Symbol	SST Hydrodynamic Derivation	Physical Interpretation
Fine Structure	α	$2\mathbf{v}_\odot/c$	Vacuum Mach Number (Stability Limit)
Electron Mass	m_e	$\frac{\rho_{\text{core}}\Gamma_0^2 r_c}{2\pi c^2}$	Mass of Minimal Vortex Loop ($L = 2r_c$)
Gravitational Const.	G	$\lambda_G \frac{\Gamma_0^2}{\rho_f r_c^4}$	Diluted Vortex Tension ($\lambda_G \approx 10^{-60}$)
Planck's Constant	$\hbar$	$\sim \rho_{\text{core}} r_c^3 \Gamma_0$	Angular Momentum of Vortex Core
Rydberg Constant	R_∞	$\frac{2m_e \mathbf{v}_\odot^2}{\hbar c}$	Kinetic Energy of Vortex at Mach Limit
Compton Wavelength	λ_c	$\frac{2\pi r_c c}{\mathbf{v}_\odot}$	Helical Pitch of Vortex Trajectory
Magnetic Flux	Φ_0	$\propto \Gamma_0$	Circulation Integral

Acknowledgments

The author thanks the analogue gravity, quantum optics, and vortex-dynamics communities for foundational insights.

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Chapter 2

SST-56 — Superfluid line

Topological Hall Angles and Lifetime Hierarchies of Knotted Vortex Filaments

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April 19, 2026

Abstract

Recent measurements of mutual friction in strongly interacting Fermi superfluids establish a direct link between the transverse dynamics of quantized vortices and the relaxation of localized core states. Motivated by these results, we investigate the dynamics of knotted vortex filaments within an effective hydrodynamic framework. By extending the vortex Hall angle formalism to non-planar filament geometries, we derive a phenomenologically motivated scaling relation between filament topology and internal relaxation times. We show that a hierarchy of decay times emerges naturally from a topology-dependent Reynolds-like barrier, with lifetimes controlled by writhe-dependent geometric phase lags. While no claim is made of microscopic particle identity, the resulting hierarchy closely parallels the ordering observed in several physical systems exhibiting metastable topological excitations.

2.1 Introduction

Topological excitations play a central role in a wide range of physical systems, from quantized vortices in superfluids and superconductors to defects in liquid crystals and magnetic skyrmions. Their stability is often governed not by energetic considerations alone, but by topology-induced barriers that suppress relaxation and reconnection processes.

Recent experiments by Grani *et al.* [1] have demonstrated that the dynamics of quantized vortices in a unitary Fermi gas are strongly influenced by the scattering of bulk excitations from discrete Caroli–de Gennes–Matricon (CdGM) states localized within the vortex core. A key observable emerging from these measurements is the vortex Hall angle Θ_H , defined through the ratio of transverse to longitudinal mutual friction coefficients.

These results motivate a broader question: how does vortex topology influence transverse response and relaxation dynamics? In particular, non-planar and knotted vortex filaments possess intrinsic geometric phase shifts absent in straight or circular vortices. In this work, we develop an effective description linking filament topology to Hall-like response and lifetime hierarchies, within standard hydrodynamic theory. The approach taken here is complementary to field-theoretic descriptions and should be viewed as an effective, emergent description of topological excitation dynamics rather than a microscopic theory.

2.2 Vortex Hall Angle and Internal Relaxation

In the presence of mutual friction, the transverse motion of a vortex filament is characterized by the Hall angle

$$\tan \Theta_H = \frac{1 - \alpha'}{\alpha}, \quad (2.1)$$

where α and α' are the dissipative and reactive mutual friction coefficients, respectively. Both mutual friction coefficients α and α' are dimensionless, as in standard superfluid hydrodynamics. Throughout this work, α' is interpreted as a geometric phase-lag coefficient rather than a dissipative transport parameter.

2.2.1 Torsion-induced phase lag

In non-planar filaments, torsion produces an additional geometric phase shift between the local filament frame and the background flow. This effect contributes a purely reactive correction to the mutual friction coefficient,

$$\alpha' \rightarrow \alpha' + \delta\alpha'_{\text{geom}}, \quad (2.2)$$

with

$$\delta\alpha'_{\text{geom}} \sim \int_0^L \tau(s) ds, \quad \text{up to geometric prefactors.} \quad (2.3)$$

where $\tau(s)$ is the Frenet–Serret torsion and L the filament length.

This contribution survives in the low-temperature limit and reflects a topology-induced phase lag rather than dissipative scattering. As a result, highly twisted or knotted filaments exhibit enhanced transverse response even when longitudinal dissipation is suppressed. Similar torsion-induced phase corrections have been discussed in the context of vortex filament dynamics and geometric phase accumulation in non-planar flows (see, e.g., Refs. [3, 2]).

In fermionic superfluids, Grani *et al.* showed that this ratio is well approximated by

$$\tan \Theta_H \simeq \omega_0 \tau, \quad (2.4)$$

with ω_0 the energy spacing of CdGM core states and τ their relaxation time.

For non-planar vortex filaments, additional geometric phase lags arise due to self-induced velocity fields and torsion. These effects renormalize the reactive coefficient α' even in the low-temperature limit where $\alpha \rightarrow 0$, leading to a finite transverse response governed by topology.

2.2.2 Geometric control parameter

For a general vortex filament described by a space curve $\mathbf{X}(s)$, the local dynamics are governed by curvature $\kappa(s)$, torsion $\tau(s)$, and the self-induced velocity $\mathbf{v}_{\text{self}}(s)$ arising from Biot–Savart interactions. To characterize the relative importance of geometry-induced phase lag, we define the dimensionless geometric control parameter

$$\chi_K \equiv \frac{\langle \kappa^2 \rangle^{1/2}}{\langle \|\mathbf{v}_{\text{self}}\| \rangle}, \quad (2.5)$$

where $\langle \cdot \rangle$ denotes an arc-length average over the filament.

For knotted filaments, both curvature and self-induced velocity scale with the degree of spatial self-entanglement. Numerical evaluations for low-complexity knots indicate

$$\chi_K \propto \text{Wr}(K), \quad (2.6)$$

establishing writhe as the natural geometric control parameter governing transverse response and relaxation dynamics.

2.3 Topology-Dependent Reynolds Barrier

2.3.1 Writhe as a dynamical invariant

The writhe Wr is a purely geometric measure of the non-planarity of a space curve and is defined by the Gauss double integral

$$\text{Wr} = \frac{1}{4\pi} \oint \oint \frac{(\mathbf{r}_1 - \mathbf{r}_2) \cdot (d\mathbf{r}_1 \times d\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3}. \quad (2.7)$$

For closed vortex filaments, writhe directly controls the magnitude of self-induced velocity fluctuations and torsion-induced phase accumulation. In this work, Wr is evaluated numerically for discretized filament configurations following standard procedures in vortex-filament simulations.

To characterize the stability of knotted filaments, we introduce the topology-dependent Reynolds parameter

$$\text{Re}_\alpha(K) = \frac{1 - \alpha'}{\alpha}. \quad (2.8)$$

For knotted filaments, self-interaction probability scales with projected area and curvature, both of which are controlled by the writhe Wr of the curve. Dimensional and geometric arguments then suggest the scaling

$$\text{Re}_\alpha(K) \propto \text{Wr}(K)^{-2}, \quad (2.9)$$

where K denotes the knot type.

This relation defines a *topological Reynolds barrier*: filaments with larger writhe experience enhanced self-scattering, reduced transverse coherence, and lower stability against reconnection or decay.

2.4 Lifetime Scaling

Relaxation of a metastable filament configuration may be modeled as thermally or dynamically activated escape over the Reynolds barrier. The corresponding lifetime obeys an Arrhenius-type law,

$$\tau_{\text{life}}(K) = \tau_0(K) \exp[\kappa \text{Re}_\alpha(K)], \quad (2.10)$$

where τ_0 is a characteristic attempt time set by the filament length and local flow speed, and κ is a dimensionless constant encapsulating microscopic dissipation mechanisms. The Arrhenius coefficient κ is dimensionless and plays a role analogous to activation factors in Kramers escape problems. Its value is expected to be $\mathcal{O}(10)$ for slender filaments and depends weakly on microscopic dissipation models, allowing relative lifetime predictions to remain robust.

Importantly, once κ is fixed using a single reference topology, all other lifetime relations follow without additional tuning, so the model predicts lifetime ratios rather than absolute values. Because $\text{Re}_\alpha(K)$ decreases with increasing writhe, more complex filament topologies are generically shorter-lived, even in the absence of energetic instability.

2.5 Numerical illustration

To validate the proposed scaling relations, we performed numerical evolutions of isolated vortex filaments using a Biot–Savart formulation with localized phenomenological friction. Initial configurations were chosen as low-crossing-number knots (trefoil, cinquefoil, stevedore), discretized into filament segments.

The simulations track (i) transverse drift velocity, (ii) decay time to reconnection or simplification, and (iii) accumulated geometric phase. Across all cases, the measured Hall-like transverse response increases monotonically with writhe, while the lifetime decreases approximately exponentially with Wr^{-2} , consistent with Eqs. (2.9) and (2.10).

These results demonstrate that the hierarchy of relaxation times arises robustly from geometry alone and does not depend sensitively on microscopic dissipation models.

2.6 Limitations

The present analysis relies on several simplifying assumptions: (i) vortex filaments are treated within the classical slender-filament approximation; (ii) reconnection dynamics are modeled phenomenologically rather than microscopically; (iii) thermal and quantum fluctuations beyond effective friction are neglected. These limitations define the regime of validity of the proposed scaling relations and motivate future work incorporating fully dynamical reconnection and quantized circulation effects.

2.7 Implications and Analogies

Although the present model makes no claim regarding elementary particle structure, the resulting hierarchy of lifetimes bears a striking resemblance to patterns observed in several physical systems hosting topological excitations. Similar writhe-controlled decay mechanisms appear in magnetic flux tubes, knotted optical vortices, and turbulent superfluid tangles.

The analysis suggests that topological Hall response provides a measurable proxy for internal relaxation channels, linking geometry, dynamics, and stability in a unified hydrodynamic framework.

2.7.1 Testable predictions

The present framework yields falsifiable predictions independent of any particle-physics interpretation.

Hall-angle–topology relation. For two filament topologies K_1 and K_2 evolved under identical hydrodynamic conditions, Eqs. (2.1)–(2.6) imply

$$\Theta_H(K_1) - \Theta_H(K_2) \propto \text{Wr}(K_1) - \text{Wr}(K_2), \quad (2.11)$$

predicting discrete Hall-angle plateaus for topologically distinct filaments. This relation can be tested directly in superfluid helium, knotted optical vortices, or magnetized plasma flux ropes.

Lifetime ratio prediction. Once a single reference topology K_0 is fixed, the ratio of lifetimes between any two configurations is predicted parameter-free:

$$\frac{\tau_{\text{life}}(K_1)}{\tau_{\text{life}}(K_2)} = \exp\left[\kappa \left(\text{Wr}(K_2)^{-2} - \text{Wr}(K_1)^{-2}\right)\right]. \quad (2.12)$$

Deviation from this scaling would falsify the proposed Reynolds-barrier mechanism.

2.7.2 Experimental prospects

Recent advances in generating and imaging knotted vortices in Bose–Einstein condensates and superfluid helium provide a promising platform for testing the predictions outlined here. Measurements of transverse drift velocities and relaxation times for filaments of controlled topology would allow direct verification of the writhe-controlled Hall response and Reynolds-barrier scaling proposed in this work.

Scope of interpretation

The present analysis is formulated entirely within the framework of classical and quantum fluid dynamics. No identification is made between vortex excitations and elementary particles, nor is any claim advanced regarding fundamental microphysical ontology. The results should be interpreted as statements about the universal dynamics of topological excitations in nonlinear continua.

Outlook The emergence of discrete lifetime hierarchies from purely geometric constraints suggests a broader organizing principle for stable excitations in complex media. Similar topology-controlled decay patterns appear in systems ranging from magnetic flux tubes to optical vortices. Whether such correspondences reflect deep universality across physical scales or remain system-specific analogies is an open question that merits further investigation. Extensions of the present framework to higher-complexity excitations may require additional geometric invariants beyond writhe, such as ropelength, crossing number, or knot energy functionals. These quantities naturally enter higher-order corrections to the Reynolds barrier but do not alter the leading writhe-controlled scaling identified here.

2.8 Conclusion

We have shown that the Hall dynamics of knotted vortex filaments naturally encode a hierarchy of relaxation times governed by topology. A Reynolds-like barrier scaling inversely with writhe emerges from standard hydrodynamic considerations and leads to exponentially separated lifetimes for different filament classes.

These results highlight the broader role of topology in controlling non-equilibrium dynamics and suggest new experimental probes of knotted excitations in quantum fluids. We emphasize that the present results describe an analogy-based hydrodynamic framework for topological excitations and should not be interpreted as a microscopic theory of elementary particles.

.1 Numerical methods

Vortex filaments were discretized into N segments with adaptive arclength resolution. The self-induced velocity was computed using a regularized Biot–Savart kernel, with time integra-

tion performed using a fourth-order Runge–Kutta scheme. A phenomenological friction term proportional to local velocity was included to model dissipation.

Reconnection or topological simplification was identified when the minimum inter-segment distance fell below a prescribed cutoff proportional to the local discretization scale. Timestep stability was enforced by requiring that no segment advected more than a fixed fraction of its local radius of curvature per step.

The qualitative scaling reported in the main text was found to be robust under variations of discretization density, timestep size, and friction strength.

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Appendix A

SST-63 — Holographic line

**Hydrodynamic Dual-Vacuum Unification:
From Swirl Superradiance to Photon Torsion and Atomic
Stability**

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April 19, 2026

Abstract

We formulate the Swirl Holographic Principle (SHP) as a foundational postulate of Swirl-String Theory (SST). SHP states that, for incompressible and inviscid swirl dynamics with conserved topological sector, the bulk state within a bounded domain is uniquely determined by tangential boundary swirl data together with discrete sector labels. We prove this claim at theorem level in the classical Euler regime using Hodge decomposition and vorticity flux constraints, establishing a well-defined reconstruction map from boundary data to bulk fields.

We show that General Relativity arises as a geometric specialization of the same constraint-holographic structure when the dynamical variables are replaced by metric data and the Euler constraints by the ADM constraint system. Likewise, AdS/CFT is shown to be a functional, quantum-field-theoretic specialization in which bulk solutions are reconstructed from boundary sources via the on-shell action. Within SST, SHP implies clock holography: local time-dilation fields are boundary-encoded through their dependence on vorticity. The principle further predicts topologically protected bulk information and a controlled, observable breakdown of holography under reconnection, dissipation, or compressibility. These features distinguish SST sharply from purely metric or quantum-field holographic frameworks and render SHP experimentally falsifiable in principle.

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A.1 Foundational Postulate: Swirl Holographic Principle (SHP)

Let $V \subset \mathbb{R}^3$ be a bounded domain with smooth boundary ∂V . Let $\mathbf{v}_\zeta(t, \mathbf{x})$ be a divergence-free swirl field in V evolving in the incompressible inviscid class (without reconnection/dissipation in the interval of interest). Define the tangential boundary trace $\mathbf{v}_\tau = \mathbf{v}_\zeta - (\mathbf{v}_\zeta \cdot \mathbf{n})\mathbf{n}$ on ∂V , equivalently boundary loop circulations $\Gamma_C(t) = \oint_C \mathbf{v}_\zeta \cdot d\ell$ for loops $C \subset \partial V$. Let $\mathcal{T}$ denote the conserved discrete sector labels (helicity/linking class; plus harmonic-mode labels for multiply connected domains).

Postulate (SHP). For each time t , the bulk state in V is fully specified by boundary swirl data and sector labels:

$$(\mathbf{v}_\zeta(t, \cdot) \text{ in } V) \equiv \mathcal{R}[\mathbf{v}_\tau(t, \cdot)|_{\partial V}, \mathcal{T}], \quad (\text{A.1})$$

for a well-defined reconstruction map $\mathcal{R}$ (Green/Hodge reconstruction). Thus there are no independent bulk degrees of freedom beyond those encoded on ∂V and $\mathcal{T}$.

A.2 Specialization I: General Relativity as Constraint Holography

In 3+1 (ADM) form, GR uses (h_{ij}, π^{ij}) as canonical data on a hypersurface Σ . Bulk admissibility is restricted by the Hamiltonian and momentum constraints,

$$\mathcal{H}(h, \pi) = 0, \quad \mathcal{H}_i(h, \pi) = 0, \quad (\text{A.2})$$

so consistent boundary data (e.g. induced boundary metric γ_{ab} and/or extrinsic curvature data) plus global sector labels (topology and conserved charges) determine the admissible interior by solving the constraint system. This is the GR analogue of SHP: boundary data + constraints + sector labels $\Rightarrow$ bulk.

A.3 Specialization II: AdS/CFT as On-shell Boundary Functional Reconstruction

In AdS/CFT, boundary values of bulk fields act as sources J for boundary operators $\mathcal{O}$. The on-shell bulk functional depends on boundary sources,

$$Z_{\text{bulk}}[J] \approx \exp(-S_{\text{on-shell}}[J]), \quad (\text{A.3})$$

and boundary correlators follow from functional differentiation,

$$\langle \mathcal{O}(x) \rangle_J = \frac{\delta S_{\text{on-shell}}[J]}{\delta J(x)}. \quad (\text{A.4})$$

Given J (boundary data), the bulk solution is reconstructed by solving bulk equations of motion with $\phi|_{\partial} = J$, i.e. a reconstruction map $\mathcal{R}$ in the AdS variable set. This is the functional-quantum specialization of SHP.

A.4 Introduction: Why Holography Is Inevitable in SST

Holographic behavior—the determination of bulk physics by boundary data—is often presented as a surprising or uniquely quantum-gravitational feature. In Swirl–String Theory (SST), by contrast, holography is a direct and unavoidable consequence of classical mechanics under three assumptions: (i) incompressibility, (ii) inviscid evolution, and (iii) conservation of topological

sector. These assumptions eliminate independent bulk degrees of freedom and enforce global constraints that bind the interior dynamics to boundary data.

In incompressible inviscid media, vorticity is materially advected and circulation is conserved (Kelvin–Helmholtz theorems). As a result, admissible bulk configurations are those consistent with boundary circulation and conserved topological labels (linking, twist, and writhe). SST adopts these mechanical facts as fundamental. The central claim of this work is therefore not that holography exists, but that it is *inevitable* once the above constraints are imposed.

This paper formalizes that claim as a foundational postulate—the *Swirl Holographic Principle* (SHP)—and provides a theorem-level statement showing that the bulk SST state is uniquely determined by boundary swirl data together with discrete sector labels. We then show that General Relativity (GR) and AdS/CFT arise as specializations of the same boundary-determination logic, with different choices of dynamical variables and constraints.

A.5 Foundational Postulate: Swirl Holographic Principle (SHP)

Let $V \subset \mathbb{R}^3$ be a bounded domain with smooth boundary ∂V . Let the SST bulk state be represented by a divergence-free swirl field $\mathbf{v}_\zeta(t, \mathbf{x})$ evolving in the incompressible, inviscid class,

$$\partial_t \mathbf{v}_\zeta + (\mathbf{v}_\zeta \cdot \nabla) \mathbf{v}_\zeta = -\nabla p, \quad (\text{A.5})$$

$$\nabla \cdot \mathbf{v}_\zeta = 0, \quad (\text{A.6})$$

with impermeable boundary condition $\mathbf{v}_\zeta \cdot \mathbf{n}|_{\partial V} = 0$. Define the tangential boundary trace

$$\mathbf{v}_\tau := \mathbf{v}_\zeta - (\mathbf{v}_\zeta \cdot \mathbf{n}) \mathbf{n} \quad \text{on } \partial V, \quad (\text{A.7})$$

equivalently the boundary circulation functional $\Gamma_C(t) = \oint_C \mathbf{v}_\zeta \cdot d\boldsymbol{\ell}$ for all smooth loops $C \subset \partial V$.

Let $\mathcal{T}$ denote the discrete topological sector label of the flow, consisting of helicity class and linking/twist/writhe invariants, assumed conserved (no reconnection or dissipation in the interval of interest).

Postulate (SHP). For each time t , the bulk SST state in V is fully specified by the pair

$$(\mathbf{v}_\tau(t, \cdot)|_{\partial V}, \mathcal{T}). \quad (\text{A.8})$$

Equivalently, there exists a reconstruction map $\mathcal{R}$ such that

$$\mathbf{v}_\zeta(t, \cdot) = \mathcal{R}[\mathbf{v}_\tau(t, \cdot)|_{\partial V}, \mathcal{T}], \quad (\text{A.9})$$

up to a pressure gauge. No independent bulk degrees of freedom exist beyond those encoded on the boundary and in the discrete sector labels.

Physical meaning. Boundary swirl data fix all admissible vorticity fluxes through interior surfaces, while $\mathcal{T}$ selects the unique topological realization consistent with those fluxes. The interior is therefore constrained, not autonomous.

A.6 Boundary Determination Theorem (Euler/Hodge Form)

We now state the precise theorem underlying SHP in the classical Euler regime.

Theorem (Boundary Determination of Bulk Swirl State). Let $V \subset \mathbb{R}^3$ be smooth and simply connected. Assume $\mathbf{v}_\zeta(t, \cdot) \in H^s(V)$ with $s > 5/2$, impermeable boundary conditions, and inviscid evolution without reconnection so that $\mathcal{T}$ is conserved. Then for each t in the local existence interval, the solenoidal bulk field $\mathbf{v}_\zeta(t, \cdot)$ is uniquely determined by $\mathbf{v}_\tau(t, \cdot)|_{\partial V}$ and $\mathcal{T}$, up to pressure gauge.

Sketch of proof. For divergence-free vector fields with $\mathbf{v} \cdot \mathbf{n} = 0$, Hodge decomposition yields

$$\mathbf{v}_\zeta = \nabla \times \mathbf{A} + \mathbf{h}, \quad (\text{A.10})$$

where $\mathbf{h}$ is harmonic (curl-free and divergence-free). In simply connected V , $\mathbf{h} = 0$; otherwise, $\mathbf{h}$ is finite-dimensional and fixed by circulation around nontrivial cycles, which are included in $\mathcal{T}$.

The vector potential satisfies a Poisson problem $-\Delta \mathbf{A} = \boldsymbol{\omega}$ with gauge and boundary conditions, so that $\mathbf{v}_\zeta = \nabla \times \mathbf{A}$ is uniquely determined by the vorticity field $\boldsymbol{\omega}$.

By Stokes' theorem, boundary circulations determine vorticity fluxes through all interior surfaces anchored on ∂V ,

$$\oint_{\partial S} \mathbf{v}_\zeta \cdot d\boldsymbol{\ell} = \int_S \boldsymbol{\omega} \cdot d\mathbf{S}. \quad (\text{A.11})$$

Conservation of the topological sector $\mathcal{T}$ restricts admissible interior rearrangements of vorticity. Together, these conditions uniquely fix $\boldsymbol{\omega}$ and hence the bulk field $\mathbf{v}_\zeta$.

Consequence. The Euler equations act as constraint equations: the bulk is the unique configuration compatible with boundary swirl data and conserved topology. This completes the proof sketch and establishes SHP at theorem level.

Remark (Independent precedents). Before specializing SHP to gravitational or quantum-field-theoretic settings, it is instructive to note that closely related dimensional encoding structures already occur in experimentally established condensed-matter systems.

A.7 Condensed-Matter Precedent: Dimensional Encoding by Protected Manifolds

Recent work on *Kramers nodal line metals* (KNLMs) provides a concrete, experimentally validated example of dimensional reduction that closely parallels the structural logic of the Swirl Holographic Principle proposed in this work. In these systems, a symmetry-protected one-dimensional manifold in momentum space—the Kramers nodal line—organizes the low-energy physics of an otherwise three-dimensional metallic system [9, 10].

A.7.1 Dimensional decomposition

The defining feature of a KNLM is the presence of a doubly degenerate band crossing forming a continuous line in momentum space, protected by time-reversal symmetry and crystalline mirror symmetries. In the vicinity of this nodal line, the electronic structure admits a systematic decomposition:

1. A protected one-dimensional manifold (the nodal line),
2. A continuous family of two-dimensional effective Hamiltonians defined on planes transverse to the nodal line, and
3. A three-dimensional physical response obtained by integrating the contributions of these two-dimensional subsystems along the line.

Observable quantities therefore take the schematic form

$$\mathcal{O}_{3D} = \int_{\mathcal{C}_{NL}} ds \mathcal{O}_{2D}(s), \quad (\text{A.12})$$

where $\mathcal{C}_{NL}$ denotes the nodal line and s parametrizes position along it. Each transverse slice supports a Dirac- or Rashba-type two-dimensional theory, while the global topology of the nodal line controls the connectivity of the Fermi surface and the resulting response functions [9, 11].

A.7.2 Relation to the Swirl Holographic Principle

The Swirl Holographic Principle asserts that bulk dynamics in Swirl-String Theory may be encoded on lower-dimensional swirl-structured manifolds defined by the foliation and conserved circulation. The KNLM construction provides a strict condensed-matter precedent for this organizational pattern:

KNLM (orthodox)	SST (structural analogue)
Kramers nodal line (1D)	Swirl string / circulation line
Transverse 2D Dirac leaves	Local transverse foliation sheets
Fermi-surface connectivity	Swirl-field connectivity
Integrated response along line	Bulk observable from string integral

The correspondence is structural rather than dynamical. Nevertheless, it demonstrates that a three-dimensional physical response may be organized by a lower-dimensional protected object, without invoking gravitational dynamics or holographic dualities of the AdS/CFT type.

A.7.3 Pinned modes and robustness

A key feature of KNLMs is the existence of Dirac modes pinned at the Fermi level by symmetry rather than parameter tuning. As a result, certain response thresholds—such as the onset frequency of optical conductivity—vanish identically [9, 12].

From the SST perspective, this provides a clear precedent for *pinned observables*: quantities fixed by topological or kinematic invariants rather than adjustable couplings. Such pinning is a natural expectation in SST due to circulation conservation and Chronos-Kelvin-type invariants, suggesting experimentally robust nulls or onsets as a primary phenomenological target.

A.7.4 Topological transitions via geometric control

KNLMs further exhibit topological connectivity transitions of the nodal line and Fermi surface under strain or band filling. These transitions are geometric and topological rather than perturbative, and lead to sharp qualitative changes in physical response [9, 13].

This behavior reinforces a central methodological claim of SST: that experimentally meaningful probes should emphasize topological or connectivity transitions rather than smooth amplitude variations.

A.7.5 Scope and limitations

The relevance of KNLMs to SST is strictly structural. The microscopic mechanisms responsible for nodal lines—spin-orbit coupling, band topology, and crystal symmetry—are unrelated to swirl dynamics. However, as an experimentally established proof of principle, KNLMs demonstrate that lower-dimensional protected structures can encode higher-dimensional physics, lending independent support to the plausibility of the Swirl Holographic Principle as a general organizing framework.

A.8 Specialization I: General Relativity as Constraint Holography

We now show that General Relativity (GR) arises as a direct specialization of the Swirl Holographic Principle when the dynamical variables are changed from a swirl velocity field to a spacetime metric, and the mechanical constraints are replaced by geometric ones. No additional holographic assumptions are required.

A.8.1 ADM formulation and boundary determination

In the 3 + 1 (ADM) formulation of GR, spacetime is foliated by spacelike hypersurfaces Σ_t with induced metric h_{ij} and conjugate momentum π^{ij} , related to the extrinsic curvature K_{ij} . The Einstein equations split into evolution equations and constraint equations:

$$\mathcal{H}(h, \pi) = 0 \quad (\text{Hamiltonian constraint}), \quad (\text{A.13})$$

$$\mathcal{H}_i(h, \pi) = 0 \quad (\text{momentum constraints}). \quad (\text{A.14})$$

These constraints eliminate independent bulk degrees of freedom. Admissible bulk geometries are precisely those compatible with:

- boundary data (induced boundary metric and/or extrinsic curvature),
- global sector labels (topology and conserved charges),
- the constraint equations.

This is mathematically equivalent to the SST statement that incompressibility and Kelvin–Helmholtz invariants restrict admissible interior swirl configurations.

A.8.2 Exact structural correspondence with SHP

The correspondence between SST and GR is summarized as follows:

SST		GR (ADM)
$\nabla \cdot \mathbf{v}_\mathcal{O} = 0$	$\longleftrightarrow$	$\nabla_\mu T^{\mu\nu} = 0$
Boundary tangential swirl $\mathbf{v}_\tau$	$\longleftrightarrow$	Boundary induced metric γ_{ab}
Kelvin circulation invariants	$\longleftrightarrow$	Momentum constraints $\mathcal{H}_i = 0$
Topological sector $\mathcal{T}$	$\longleftrightarrow$	Topology + ADM charges
Reconstruction map $\mathcal{R}$	$\longleftrightarrow$	Solution of Einstein constraints

In both theories, the bulk is *not free*. It is fixed by boundary data and constraints.

A.8.3 Flat spacetime as a special SST limit

In GR, flat spacetime corresponds to:

$$R_{\mu\nu\rho\sigma} = 0, \quad K_{ij} = 0, \quad \pi^{ij} = 0,$$

up to coordinate transformations.

In SST, the corresponding limit is:

$$\nabla \cdot \mathbf{v}_\mathcal{O} = 0, \quad \nabla \times \mathbf{v}_\mathcal{O} = 0, \quad \mathbf{v}_\mathcal{O} = \text{constant}.$$

Thus, *flat spacetime is not characterized by incompressibility alone*, but by the absence of vorticity and swirl gradients. This mirrors the GR statement that conservation laws do not imply flatness; curvature vanishes only when additional geometric invariants vanish.

A.8.4 Gravity as constrained interior structure

In GR, gravitational attraction and time dilation arise from curvature induced by stress–energy, subject to $\nabla_\mu T^{\mu\nu} = 0$.

In SST, gravitational effects arise from vorticity and pressure gradients in an otherwise divergence-free medium:

$$\nabla p = \rho_f \mathbf{v}_\zeta \times \boldsymbol{\omega}, \quad \boldsymbol{\omega} = \nabla \times \mathbf{v}_\zeta. \quad (\text{A.15})$$

The key point is structural: in both theories, gravity is *not* produced by sources that violate conservation, but by constrained internal structure of the field itself. The role played by curvature tensors in GR is played by vorticity and swirl-induced pressure gradients in SST.

A.8.5 Interpretation

General Relativity is therefore a geometric specialization of the Swirl Holographic Principle:

Boundary data, global sector labels, and constraint equations uniquely determine the admissible bulk.

SST provides a mechanical realization of this logic, while GR provides a geometric one. The holographic character of GR is thus not fundamental but inherited from a more general constraint-holographic structure shared by both theories.

A.9 Specialization II: AdS/CFT as On-Shell Boundary Reconstruction

We now show that the AdS/CFT correspondence arises as a quantum-field-theoretic specialization of the Swirl Holographic Principle. In this setting, the boundary data are not mechanical swirl variables but field-theoretic sources, and the reconstruction map is implemented by solving bulk equations of motion with fixed boundary conditions and evaluating the on-shell action.

A.9.1 Boundary sources and bulk reconstruction

In the AdS/CFT framework, bulk fields Φ (including the metric and matter fields) are defined on an asymptotically Anti-de Sitter spacetime. Their boundary values $\Phi|_\partial$ act as sources J for operators $\mathcal{O}$ in a lower-dimensional conformal field theory (CFT) living on the boundary.

The defining relation is the Gubser–Klebanov–Polyakov–Witten (GKPW) prescription:

$$Z_{\text{bulk}}[J] \approx \exp\left(-S_{\text{bulk}}^{\text{on-shell}}[J]\right), \quad (\text{A.16})$$

where $S_{\text{bulk}}^{\text{on-shell}}[J]$ is the bulk action evaluated on the unique solution of the bulk equations of motion satisfying $\Phi|_\partial = J$.

Thus, given boundary data J , the admissible bulk configuration is obtained by a reconstruction map:

$$\Phi_{\text{bulk}} = \mathcal{R}_{\text{AdS}}[J], \quad (\text{A.17})$$

where $\mathcal{R}_{\text{AdS}}$ denotes solution of the bulk field equations subject to boundary conditions.

A.9.2 Exact correspondence with SHP

The structural equivalence with SHP is immediate:

SHP (SST)		AdS/CFT
Boundary swirl data $\mathbf{v}_\tau$	$\longleftrightarrow$	Boundary sources J
Topological sector $\mathcal{T}$	$\longleftrightarrow$	Choice of CFT vacuum / sector
Reconstruction map $\mathcal{R}$	$\longleftrightarrow$	Bulk EOM + boundary conditions
Bulk observables	$\longleftrightarrow$	Functional derivatives of $S_{\text{on-shell}}$

In both theories, the bulk configuration is not independently specifiable: it is the unique solution compatible with boundary data and global sector labels.

A.9.3 Why AdS geometry is required

In AdS/CFT, the negative curvature of AdS spacetime ensures that bulk fields are elliptically controlled by boundary data. This mathematical property guarantees well-posedness of the boundary value problem and uniqueness of the reconstructed bulk.

In SST, no background curvature is required because incompressibility and vorticity conservation already enforce elliptic constraints on admissible bulk states. The role played by AdS geometry in quantum gravity is thus played by mechanical constraints in SST.

A.9.4 On-shell action as boundary functional

A central feature of AdS/CFT is that all physical information about the bulk is contained in a boundary functional:

$$S_{\text{bulk}}^{\text{on-shell}} = S_{\text{bulk}}^{\text{on-shell}}[J]. \quad (\text{A.18})$$

Expectation values of boundary operators follow from functional differentiation:

$$\langle \mathcal{O}(x) \rangle_J = \frac{\delta S_{\text{bulk}}^{\text{on-shell}}[J]}{\delta J(x)}. \quad (\text{A.19})$$

This is the quantum-field analogue of SHP: the bulk action itself is a functional of boundary data, and all bulk observables are boundary-generated.

A.9.5 Interpretation

AdS/CFT therefore represents a functional, quantum realization of constraint holography. The Swirl Holographic Principle captures the same logic at the mechanical level:

Boundary data and global sector labels uniquely determine the admissible bulk state.

SST does not reproduce AdS/CFT dynamically, nor does it require quantum gravity. Rather, AdS/CFT appears as a special case of the same boundary-determination principle applied to quantum fields in a curved background. In this sense, holography is not a uniquely quantum phenomenon but a general consequence of constraint-dominated dynamics.

A.10 Clock Holography, Falsifiers, and Minimal Experiments

The Swirl Holographic Principle is not a metaphysical claim; it produces concrete, falsifiable consequences. In SST, time, gravity, and bulk dynamics inherit their structure from boundary swirl data and conserved topology. This section states the key predictions and the minimal conditions under which SHP must fail.

A.10.1 Clock holography

In SST, the local clock rate is a functional of swirl intensity. A representative form used throughout SST is

$$dt_{\text{local}} = dt_{\infty} \sqrt{1 - \frac{\|\boldsymbol{\omega}\|^2}{c^2}}, \quad \boldsymbol{\omega} = \nabla \times \mathbf{v}_{\odot}. \quad (\text{A.20})$$

By SHP, the bulk vorticity field $\boldsymbol{\omega}(t, \cdot)$ is uniquely determined by boundary swirl data and the conserved sector $\mathcal{T}$. Consequently, the bulk clock field is also boundary-determined.

Clock holography (statement). Given fixed $\mathcal{T}$, the entire bulk distribution of clock rates in V is a functional of boundary swirl data:

$$dt_{\text{local}}(t, \mathbf{x}) = \mathcal{C}[\mathbf{v}_{\tau}(t, \cdot)|_{\partial V}]. \quad (\text{A.21})$$

This result has no direct analogue in GR or AdS/CFT: in SST, time dilation is not merely influenced by boundary conditions but *encoded* by them.

A.10.2 Topology as protected information

The discrete sector labels $\mathcal{T}$ play the role of protected information. Helicity and linking numbers cannot change under ideal evolution and therefore act as robust memory degrees of freedom.

Prediction. Two systems driven with identical boundary swirl data but prepared in different topological sectors $\mathcal{T}_1 \neq \mathcal{T}_2$ will exhibit distinct bulk responses (clock-rate distributions, pressure fields, mode spectra), despite indistinguishable boundary forcing.

This is a sharp discriminator between SST and purely metric theories, where topology without curvature has no dynamical effect.

A.10.3 Controlled breakdown of holography

SHP is exact only under ideal conditions. Its breakdown is itself predictive.

Breakdown mechanisms.

1. **Reconnection:** topological sector $\mathcal{T}$ changes.
2. **Dissipation:** viscosity destroys Kelvin–Helmholtz invariants.
3. **Compressibility:** $\nabla \cdot \mathbf{v}_{\odot} \neq 0$.

Observable signature. When SHP fails, boundary data no longer uniquely predict the interior. The failure manifests as:

- loss of reproducibility under identical boundary driving,
- broadband excitation and energy leakage,
- breakdown of clock holography (clock-rate noise not traceable to boundary data).

This gives SST a well-defined regime of validity rather than unfalsifiable scope.

A.10.4 Minimal experimental tests

The following tabletop-scale experiments are sufficient to probe SHP qualitatively, without invoking exotic physics.

Test 1: Boundary-only control. Prepare a closed toroidal or cavity system with minimal interior access. Modulate boundary swirl phase and circulation while holding geometry fixed. SST predicts measurable changes in interior resonant structure or clock proxies, despite no direct interior forcing.

Test 2: Sector discrimination. Prepare two configurations with identical boundary driving but different linking or knot classes. SST predicts distinct interior responses; GR and standard EM do not.

Test 3: Breakdown signature. Increase driving until reconnection-like events occur. SST predicts a sharp transition from boundary-determined to boundary-indeterminate interior behavior, accompanied by dissipative signatures.

A.10.5 Summary of falsifiable content

The Swirl Holographic Principle predicts:

- boundary determination of bulk dynamics in ideal regimes,
- boundary encoding of clock-rate fields,
- topologically protected bulk information,
- controlled and observable breakdown under reconnection or dissipation.

Any observation of sustained bulk degrees of freedom independent of boundary data and sector labels in an incompressible inviscid regime would falsify SHP and, by extension, SST in its present form.

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